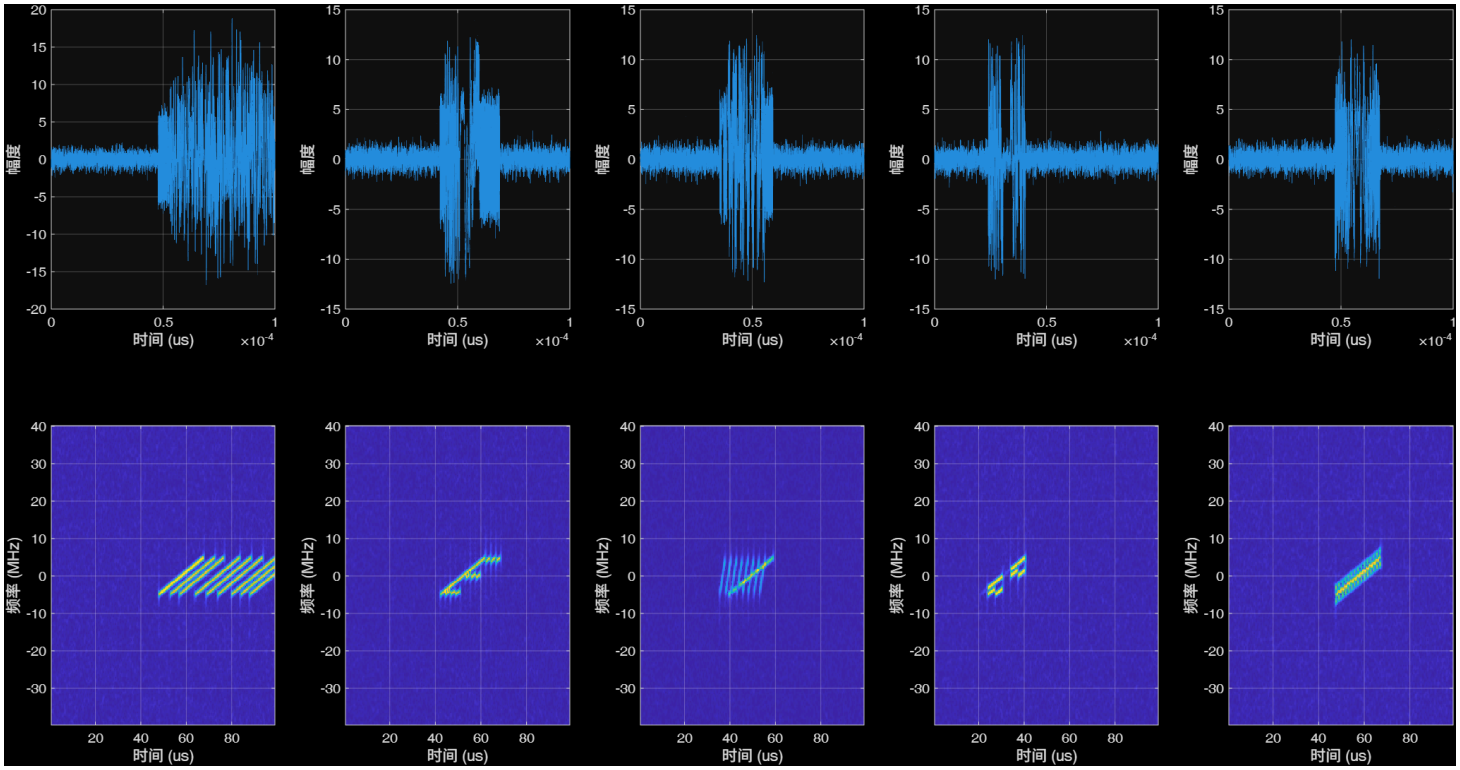
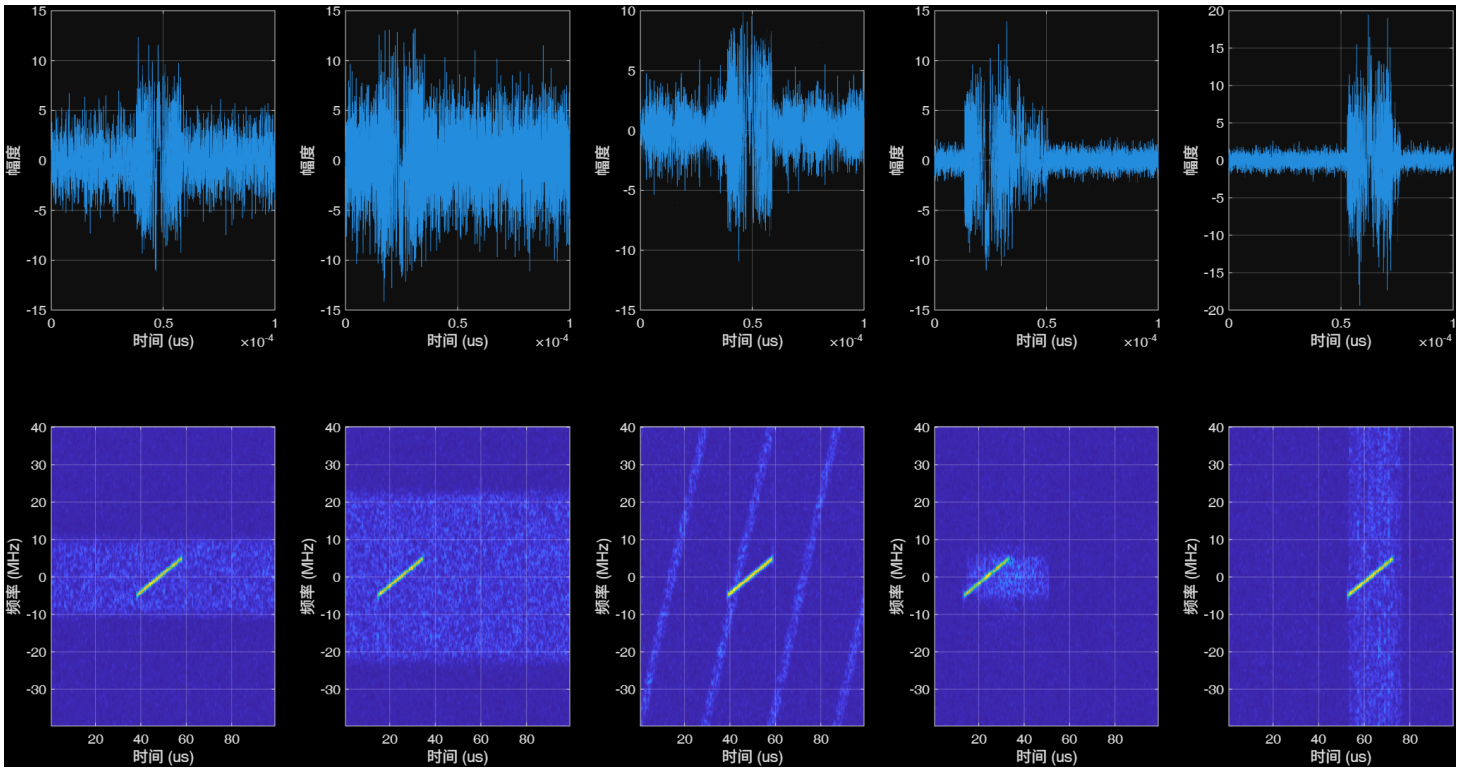
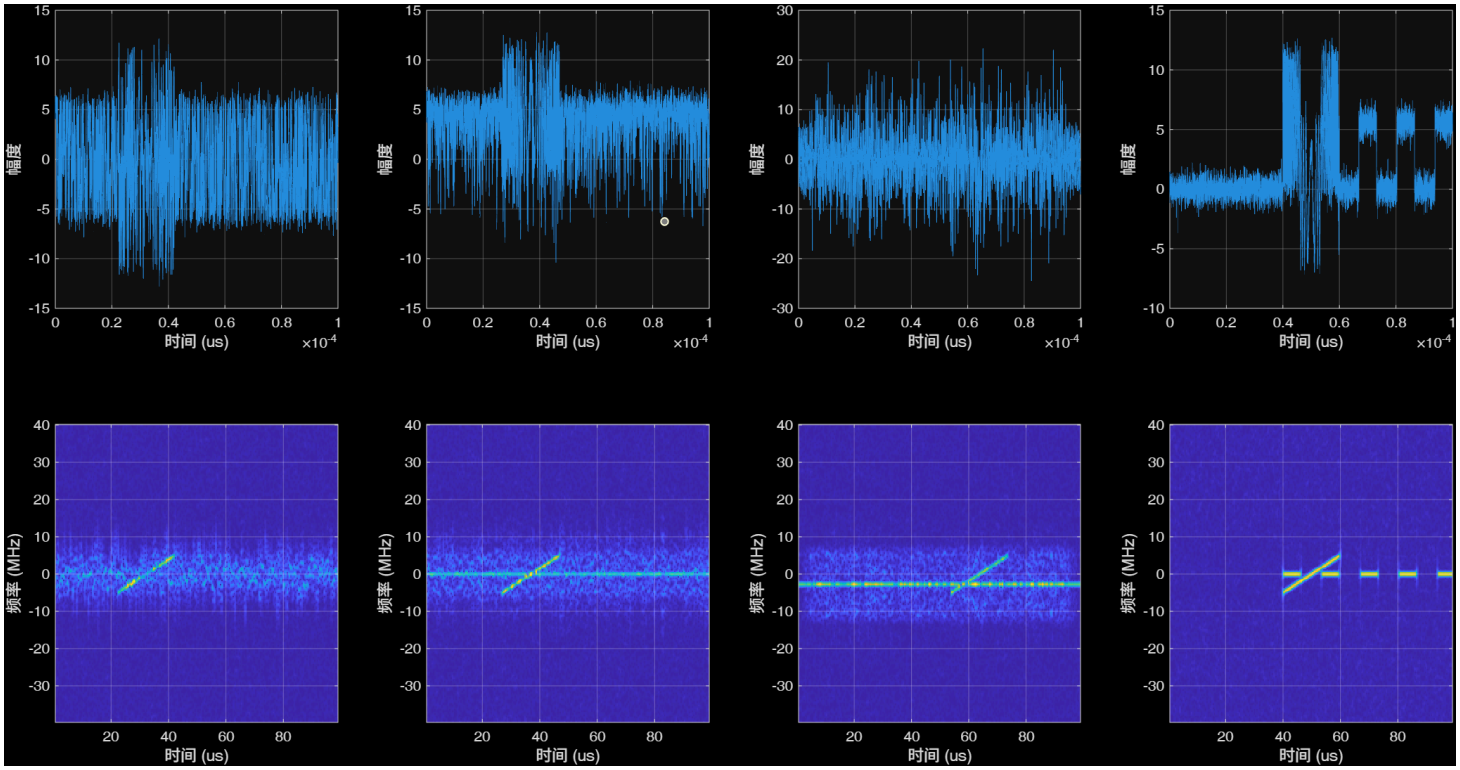
组合零样本小测试

欺骗+压制+组合3+3+9







train = ['DFTJ', 'ISRJ', 'SMSPJ', 'AJ', 'BJ','SJ']
test = ['DFTJ', 'ISRJ', 'SMSPJ', 'AJ', 'BJ','SJ']
val = ['DFTJ', 'ISRJ', 'SMSPJ', 'AJ', 'BJ','SJ'+九种组合]

设定为主瓣干扰，

样本个数:train 每个种类210个样本,val 90个,test 600个
7:3:20的比例
SNR = 15 dB，JNR = 0，10，20, 30，40, 50 dB
(目标信噪比是否需要一定的变化?)

HDF5 -v7.3保存
xx_echo_label.mat 多热标签
xx_echo_stfts.mat 时频域数据 每张图大小为124x64，complex single
xx_echo_times.mat 时域数据 每段回波长度为1x8000，complex single

self.classes = [
'DFTJ', # 密集假目标干扰
'ISRJ', # 间歇采样转发干扰
'VDJ', # 速度欺骗干扰
'DDJ', # 距离-速度联合欺骗干扰
'AJ', # 瞄准干扰
'BJ', # 阻塞干扰

'SJ', # 扫频干扰
'NCJ', # 噪声卷积干扰
'NPJ', # 噪声乘积干扰
'SPSMJ', # 弥散谱干扰
'C&IJ', # 切片交织干扰
'NFMJ', # 噪声频率调制干扰
'NPMJ', # 噪声相位调制干扰
'NAMJ', # 噪声幅度调制干扰
'CSJ', # 梳状谱干扰
'PJ' # 脉冲干扰
]

每个类别的文本描述（用于零样本学习）

self.descriptions = {
'DFTJ': [
'a radar signal with dense false target jamming',
'a radar echo with multiple delayed and stacked false targets',
'a radar signal intercepted, delayed and retransmitted many times to create false targets'
],
'ISRJ': [
'a radar signal with interrupted sampling repeater jamming',
'a radar echo with periodic sampled and forwarded interference',
'a radar signal with interference sampling and immediate forwarding jamming'
],
'VDJ': [
'a radar signal with velocity deception jamming',
'a radar echo with false Doppler frequency modulation',
'a radar signal with velocity drag deception interference'
],
'DDJ': [
'a radar signal with distance and velocity joint deception jamming',
'a radar echo with combined range and Doppler deception',
'a radar signal with simultaneous range-velocity deception interference'
],
'AJ': [
'a radar signal with aiming suppression jamming',
'a radar echo with frequency-aligned narrowband jamming',

```
'a radar signal with targeted frequency coverage jamming'
],
'BJ': [
'a radar signal with barrage suppression jamming',
'a radar echo with wideband barrage interference',
'a radar signal with broadband noise covering multiple frequencies'
],
'SJ': [
'a radar signal with swept frequency jamming',
'a radar echo with periodically varying center frequency interference',
'a radar signal with sweeping carrier frequency jamming'
],
'NCJ': [
'a radar signal with noise convolution jamming',
'a radar echo with convolved gaussian noise interference',
'a radar signal with noise convolution smart jamming'
],
'NPJ': [
'a radar signal with noise multiplication jamming',
'a radar echo with multiplied gaussian noise interference',
'a radar signal with noise product smart jamming'
],
'SPSMJ': [
'a radar signal with smeared spectrum jamming',
'a radar echo with spread spectrum interference from sub-pulses',
'a radar signal with interval-sampled and compressed sub-pulse jamming'
],
'C&IJ': [
'a radar signal with chopping and interleaved jamming',
'a radar echo with sliced and reconstructed interference',
'a radar signal with time-sliced and repeatedly filled jamming'
],
'NFMJ': [
'a radar signal with noise frequency modulation jamming',
'a radar echo with instantaneous frequency modulated by noise',
'a radar signal with noise-modulated instantaneous frequency'
],
'NPMJ': [
```

```
'a radar signal with noise phase modulation jamming',
'a radar echo with instantaneous phase modulated by noise',
'a radar signal with noise-modulated instantaneous phase'
],
'NAMJ': [
'a radar signal with noise amplitude modulation jamming',
'a radar echo with amplitude modulated by gaussian noise',
'a radar signal with noise-modulated signal amplitude'
],
'CSJ': [
'a radar signal with comb spectrum jamming',
'a radar echo with comb-shaped frequency distribution',
'a radar signal with multiple false targets from multi-frequency modulation'
],
'PJ': [
'a radar signal with pulse jamming',
'a radar echo with rectangular pulse interference on carrier',
'a radar signal with repetitive high-frequency pulse jamming'
]
}
```